



Italo Baros

MSc. in CS | Data & AI Architect | Databricks Certified | GCP Certified | Azure | Python Expert

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🌐 <https://www.linkedin.com/in/ircbarros/> 🐙 <https://github.com/Ircbarros>
📄 <https://stackoverflow.com/users/11595480/ircbarros>

Skills

☁ Microsoft Azure

Azure Blob Storage, Azure Data Lake, Azure Event Hub, Azure Databricks, Azure Data Explorer, Azure Data Factory, Microsoft Copilot, PowerBI, Azure Key Vault, Azure IoT Hyb, Azure Power App

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⚙ DataSecOps

DataOps, Data Quality, MLOps, Data Governance, Data Security, GenAI, Machine Learning

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☁ Google Cloud Platform

BigQuery, Dataflow, PubSub, Databricks, Cloud Functions, Cloud Storage, Compute Engine, Container Registry, Cloud Interconnect, Cloud Virtual Network

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📄 Data Engineering

Python, Airflow, Pandas, Object-Oriented Programming (OOP), Unit Testing, Prefect, Data Modeling, Git, ETL, Streaming, SQL, NoSQL

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📊 Databricks

ACLs, FGA, Administration, Liquid Clustering, PySpark, Machine Learning, MLOps, Spark SQL, Asset Bundles, Unity Catalog

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🔧 Additional Tools

Kubernetes, Terraform, Docker, Linux Administration, Networking, Containers, Open Telemetry

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Experience

EPAM Systems \ ASML

Lead Data Engineer

Full Remote
March, 2025 - Present

- Improved and maintained a multi-cloud Delta Sharing solution at GB/TB scale on Databricks, enabling seamless, bidirectional data synchronization between Azure and GCP via Unity Catalog, Delta Sharing, Asset Bundles, and FGA.
- Ensured accurate synchronization of source machine data between Azure and GCP for Vertex AI use-case teams focused on detecting machine faults and optimizing operational efficiency and diagnostics.
- Debugged and resolved infrastructure and networking issues in GCP related to the Customer Managed VPCs, Firewall Rules, NGINX, PSCs, Load Balancers, Cloud Interconnect, and Databricks Back-end.
- Developed a solution to aggregate Databricks logs across multiple workspaces on a centralized platform using Kubernetes, OpenTelemetry, and Azure IoT Hub.

Devoteam DCloud (Previous GCloud)

Data Architect

Full Remote
March, 2024 - March, 2025

- Developed scalable data architectures for multinational companies in the sectors of telecommunications, retailers, and 4.0 industries (IoT), also participating in the Pre-Sales phases.
- Responsible to leverage Google Cloud services and designing the clients Data Architecture following the DAMA and Google best practices.
- Responsible for building and managing PoCs and MVPs using GCP Data Services based on the business and stakeholders' requirements.
- Actively collaborates with Data Engineers, Architects, and Stakeholders to design secure, reliable, and cost-effective solutions for Big Data scenarios, including IoT, On-Prem to Cloud migration. Systems Integration and Architecture Revamp.
- Built and improved the Technical Design document containing Naming Conventions, Guidelines, and Best Practices for multiple scenarios (e.g. disaster recovery, data privacy, data modeling) to make the implementations consistent across clients and projects.

Experience

Aubay Portugal \ Beyond Gravity
Data Management Specialist

Lisbon, Portugal (Hybrid)
November, 2023 - March, 2025

- Responsible for guiding, leveraging, and implementing the base foundation for Beyond Gravity's future Data Organization based on DAMA principles.
- Implement the BG's Data Organization plan using inputs from Legal, Compliance, Business, and IT teams, showing the steps needed to achieve a state-of-the art Data Security, Data Quality, Data Governance, and Management programs.
- Provided support to the legacy systems migration to a cloud-oriented approach, developing the enterprise ERD and Data Model.
- Built a GenAI Data Catalog and Business Glossary product using Microsoft Copilot Studio and GPT to help users to migrate easily into the new architecture.

Aubay Portugal \ Reckitt Benckiser
Data Engineer

Lisbon, Portugal (Hybrid)
March, 2022 - October, 2023

- DataSecOps Tech Leader, helping to conduct PoCs and implement trending technologies using IA, LLMs, and Open Source Libraries to reduce Cloud Costs and Improve Data Quality & Governance.
- Helped to migrate the Legacy Platform into the Modern Data Stack, working closely with the Business Users during the data validation.
- Translated Legacy SQL code and SQL SPs into PySpark, reducing the ETL transformation from hours to minutes.
- Helped and advised the Data Architecture team to review future platform new implementations, migrations, and integrations with tools like Databricks, SODA, DBT, Great Expectations, Unravel, Acceldata, e.g.

AES Brasil
Analyst

Full Remote
September, 2020 - February, 2022

- Responsible for the data architecture design, implementation, and collection used to run AI and ML models for predicting factors such as Wind Energy Generation, System Load, Flows Forecasts, and Rain Forecasts used to make buy/sell decisions. Reduced the manual work from a full day to 4 hours.
- Lead Data Engineer responsible for understanding and translating the Business Users' Requests into Automated ETLs & Near-Real Time Reports. Reduced the delivery to the business users from 3 hours to 20 minutes.
- Worked closely with third parties and consultancy companies in joint Machine Learning projects, providing advisory on the ETL, Automation, MLOps, and Web-Scraping processes.
- Provided training to the AES Corporation's main Data Engineer team on workflow management technologies, such as Prefect.

Education

Federal University of Paraiba (UFPB)
Informatics

MSc.
Paraiba, Brazil • March, 2019 - April, 2021

Federal University of Semi-Arid Region (UFERSA)
Electrical Engineering

B. Eng.
Rio Grande do Norte, Brazil • February, 2016 - October, 2018

Federal University of Semi-Arid Region (UFERSA)
Applied Engineering

B. Sc.
Rio Grande do Norte, Brazil • February, 2012 - December, 2015

Languages

English

Portuguese

Spanish

Russian

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Certifications

<u>Databricks Certified Data Engineer Professional</u> Databricks	October, 2025
<u>Professional Cloud Database Engineer Certification</u> Google Cloud	April, 2025
<u>SI Associate Certification</u> MongoDB	January, 2025
<u>SI Architect Certification</u> MongoDB	January, 2025
<u>Associate Data Practitioner Certification</u> Google Cloud	November, 2024
<u>Prefect Associate</u> Prefect	September, 2023
<u>Databricks Certified Data Engineer Associate</u> Databricks	July, 2023
<u>Astronomer Certification for Apache Airflow Fundamentals</u> Astronomer	May, 2023
<u>dbt Fundamentals</u> DBT https://www.credential.net/3405e4b4-9eb2-4eb0-929f-96063b6b5c2f	March, 2023
<u>DevOps Essentials Professional Certificate (DEPC)</u> Certiprof	August, 2020
<u>API Designer</u> API Academy https://1drv.ms/b/s!Amtyl-5s6u0j02g31wSb0fvVfSvq?e=elyvtb	August, 2020

Publications

<u>Robotic Mobile Fulfillment Systems: A survey on recent developments and research opportunities</u> Robotics and Autonomous Systems Journal	March, 2021
<u>TurtleUI: A Generic Graphical User Interface for Robot Control</u> Latin American Robotics Symposium (LARS)	September, 2019